

VERIFICATION OF GATES

Aim:

To study and verify the truth table of logic gates

Apparatus required:

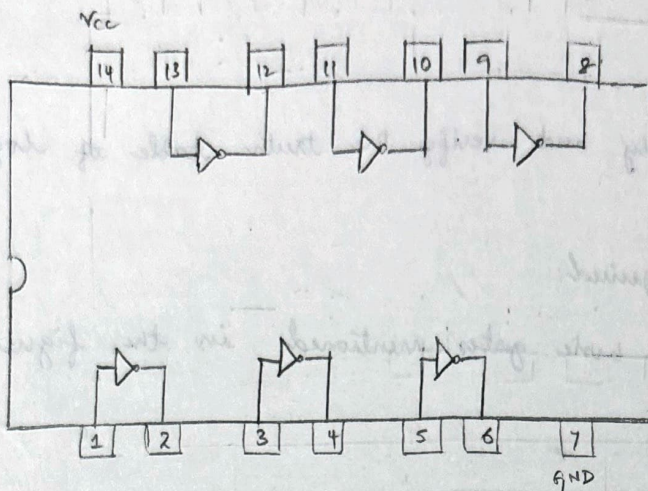
All the basic gates mentioned in the figure.

Procedure:

1. Place the IC on the IC trainer kit
2. Connect V_{cc} and ground to respective pins to the IC trainer kit.
3. Connect the inputs to the input switches provided in the IC trainer kit.
4. Connect the outputs to the switches of O/P LEDs.
5. Apply various combinations of inputs according to the truth and observe the condition of LEDs.
6. Disconnect output from the LEDs and note down the corresponding multimeter voltage reading for various combinations of inputs.

Inverter gate (NOT gate)

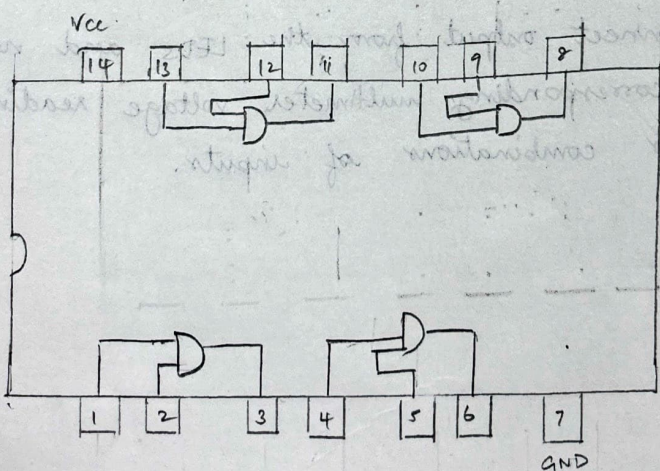
7404 LS



A	Output	Y_1 (V)	Y_2 (V)	Y_3 (V)	Y_4 (V)	Y_5 (V)	Y_6 (V)
0	1	5V	5V	5V	5V	5V	5V
1	0	0V	0V	0V	0V	0V	0V

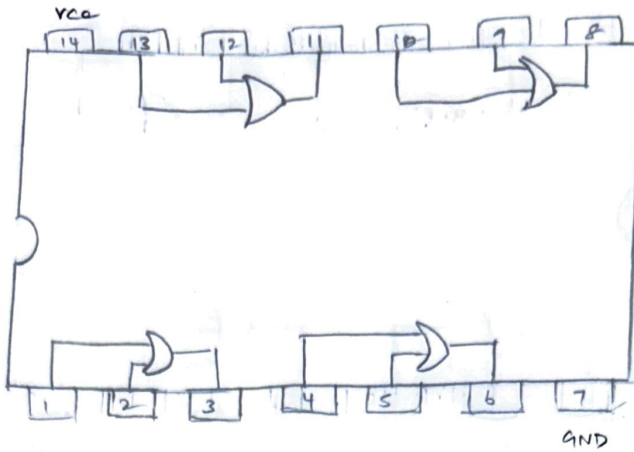
2- Input AND gate

7408 LS



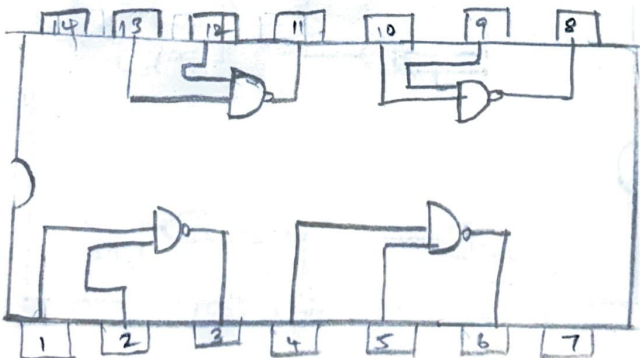
A	B	Output	Y_1 (V)	Y_2 (V)	Y_3 (V)	Y_4 (V)
0	0	0	0V	0V	0V	0V
0	1	0	0V	0V	0V	0V
1	0	0	0V	0V	0V	0V
1	1	1	5V	5V	5V	5V

Input OR gate — 7432 LS —



A	B	output	$V_1(V)$	$V_2(V)$	$V_3(V)$	$V_4(V)$
0	0	0	0V	0V	0V	0V
0	1	1	5V	5V	5V	5V
1	0	1	5V	5V	5V	5V
1	1	1	5V	5V	5V	5V

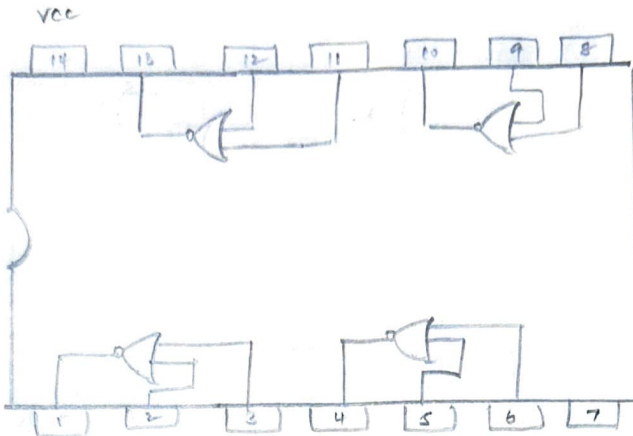
Input NAND gate — 7400 LS —



A	B	O/P	$V_1(V)$	$V_2(V)$	$V_3(V)$	$V_4(V)$
0	0	1	5V	5V	5V	5V
0	1	1	5V	5V	5V	5V
1	0	1	5V	5V	5V	5V
1	1	0	0V	0V	0V	0V

Input NOR gate

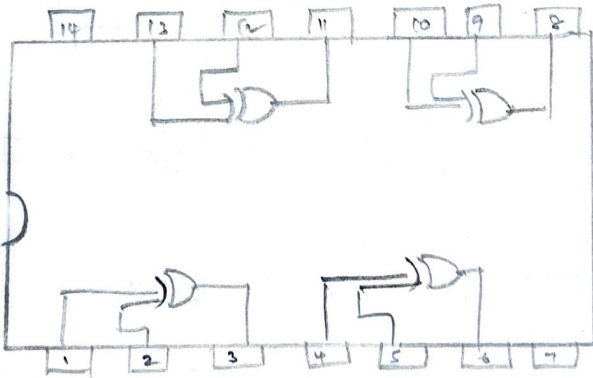
7402 LS



A	B	O/P	V_1 (V)	V_2 (V)	V_3 (V)	V_4 (V)
0	0	1	5V	5	5	5
0	1	0	0	0	0	0
1	0	0	0	0	0	0
1	1	0	0	0	0	0

Input EX-OR gate

7486 LS



A	B	O/P	V_1 (V)	V_2 (V)	V_3 (V)	V_4 (V)
0	0	0	0V	0V	0V	0V
0	1	1	5	5	5	5
1	0	1	5	5	5	5
1	1	0	0	0	0	0

CONCLUSION:

Thus, the AND, OR, NOT, NAND, NOR, and XOR gates were successfully verified during respective ICs.